

JKU
Visual Data
Science Lab

AI and Visualization

Fundamentals and Explaining Algorithms

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Recap of Lecture 1

Introduction
and
Course Overview

- Course Overview
- Introduction
- Overview of Topics

Lecture 2

Fundamentals
and
Explaining Algorithms

- AI & VIS Fundamentals

- Explaining Sorting Algorithms
- Explaining Clustering Algorithms & Results
- Explaining Bayes' Theorem
- Explaining Decision Trees

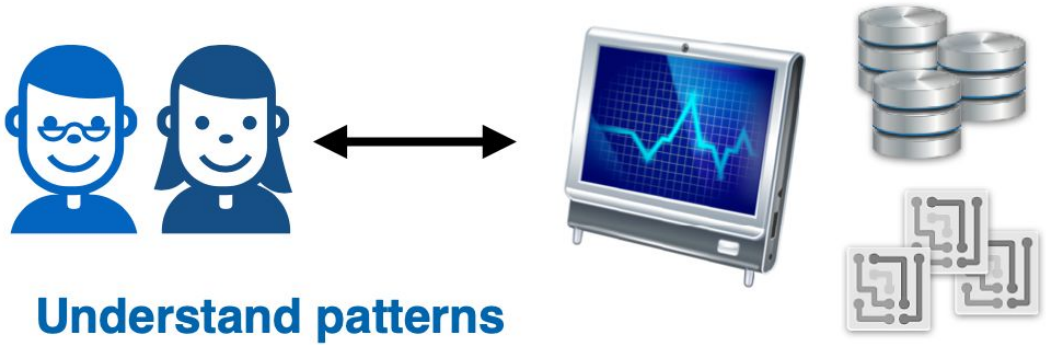
AI & VIS Fundamentals

AI + VISUALIZATION + HUMAN = EXPLAINABLE AI



?

Human-Data Interaction



Understand patterns

Gain insights

Make decisions

Communicate

Why Visualization?

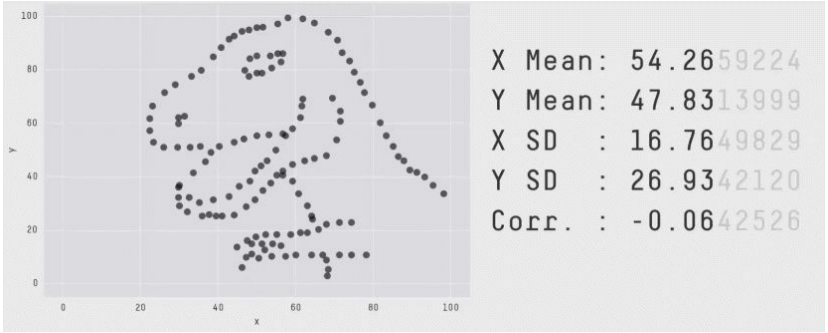
Human can see patterns that algorithms cannot

1		2		3		4	
X	Y	X	Y	X	Y	X	Y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

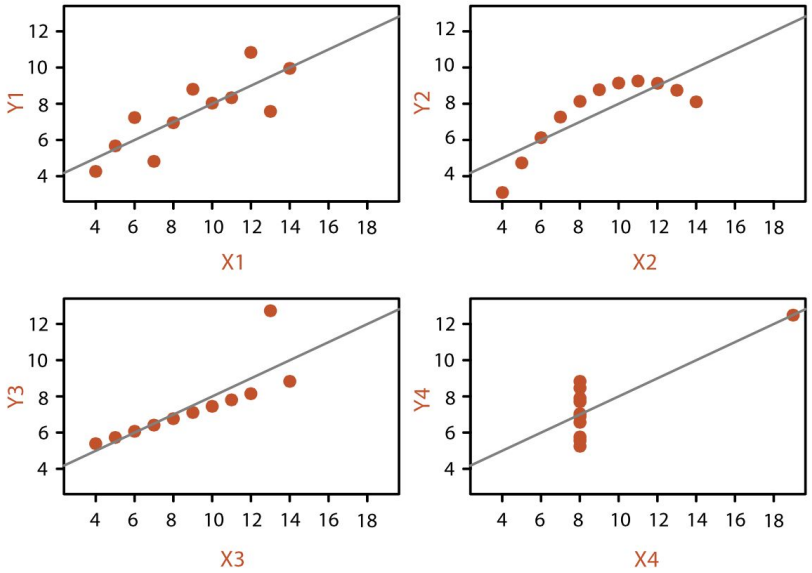


Francis Anscombe

Anscombe's Quartett Anscombe 1973




```
X Mean: 9.00
Y Mean: 7.50
X SD  : 3.31
Y SD  : 2.03
Corr. : 0.81
```



Matejka & Fitzmaurice, 2017. doi :10.1145/3025453.3025912

Definition: Visualization (Vis)

> Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.
>
> *Tamara Munzner*

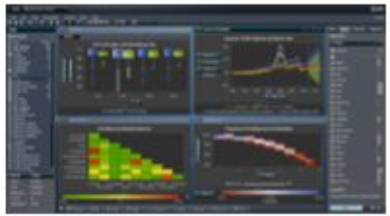
Interactive Visual Analysis

Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.

Tamara Munzner



Tableau



SAS



Improvise



IBM Watson Analytics

Not pretty pictures!

Why a Human in the Loop?

> Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.
>
> *Tamara Munzner*

Not needed if fully automatic solution exists (and is trusted)

Needed for ill-defined / ill-structured problems

- No single optimal solution
- No clear objective measures (example: find cure for cancer)

Examples

- Exploratory analysis of scientific problems
- (Collaborative) decision-making problems
- Model building & validation

Effective Support of Ill-defined Tasks

> Computer-based visualization systems provide visual representations of datasets designed to help people carry out **tasks** more effectively.
>
> *Tamara Munzner*

Metrics of Success

- Novel — entire new insights
- Faster — speed up common workflows

Why Visual Representations?

Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.

Tamara Munzner

Triglyceride levels

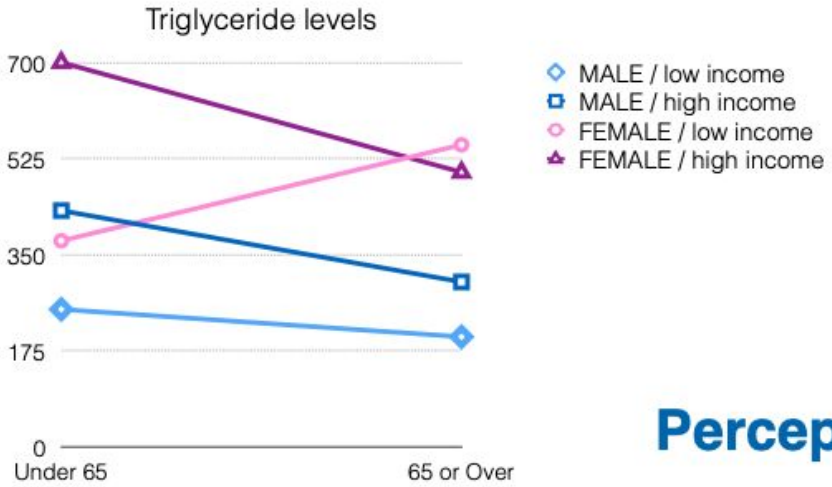
	Males		Females	
Income Group	**Under 65**	**65 or Over**	**Under 65**	**65 or Over**
low income	250	200	375	550
high income	430	300	700	500

Q: Which gender or income level group shows different effects of age on triglyceride levels?

Why Visual Representations?

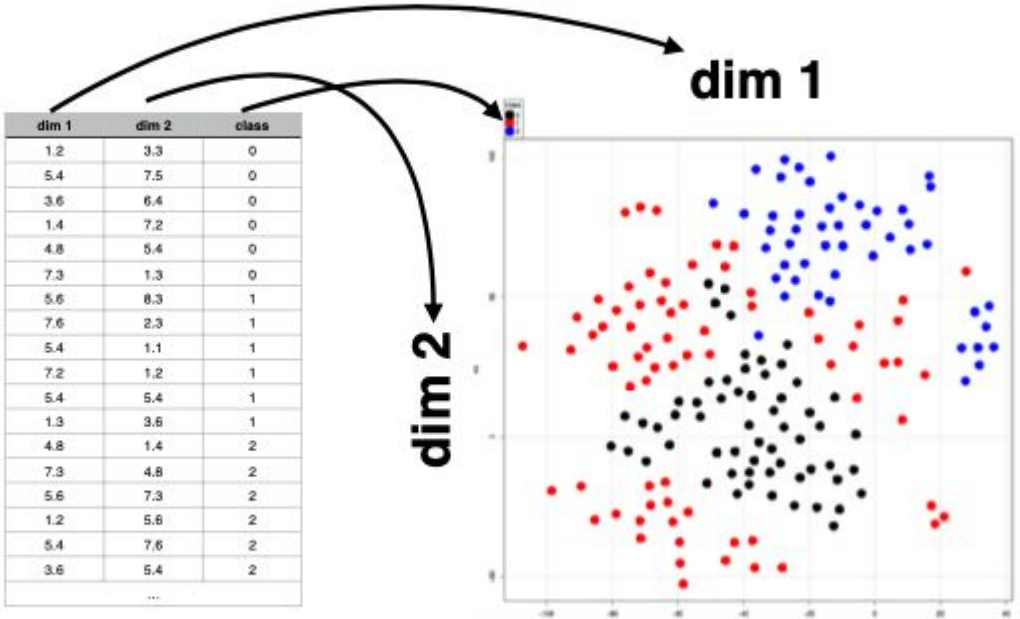
Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.

Tamara Munzner



Perception beats Cognition!

Visual Encoding



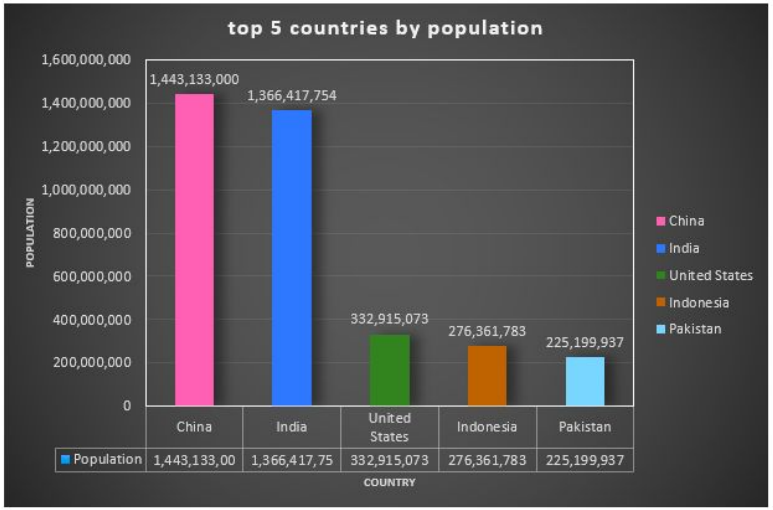
Data

Visualization

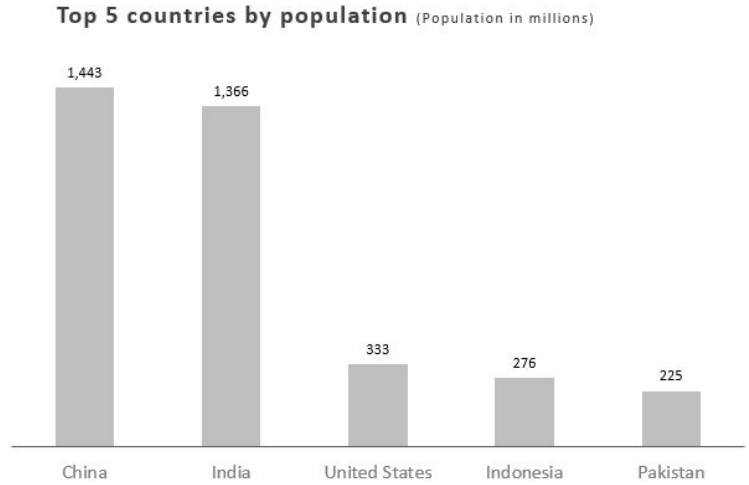
Visual Design

$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$

Before



After

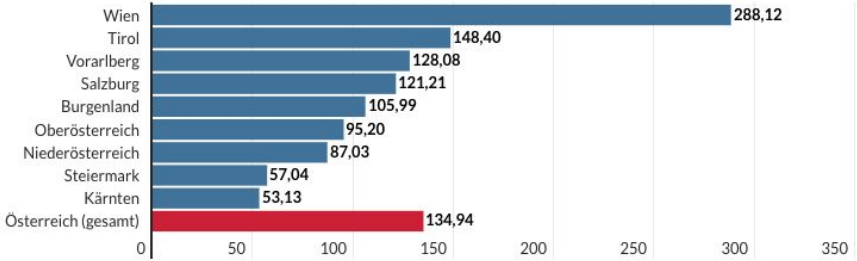


https://www.linkedin.com/pulse/improving-data-visualization-using-principle-data-ink-wijesinghe/

Edward Tufte

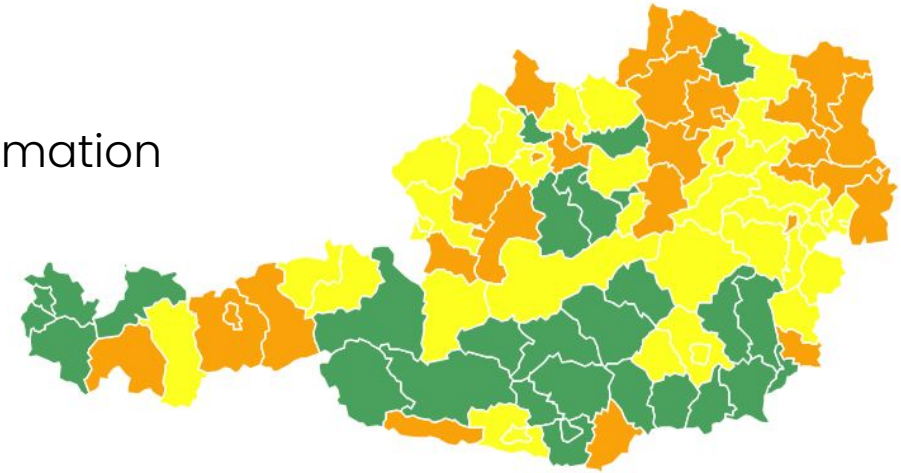
What We Can Do Now…

Static representations to present information



Relative Corona Cases in Austria

BMSGPK



Corona Traffic Light Map

BMSGPK



Tamara Munzner



Colin Ware



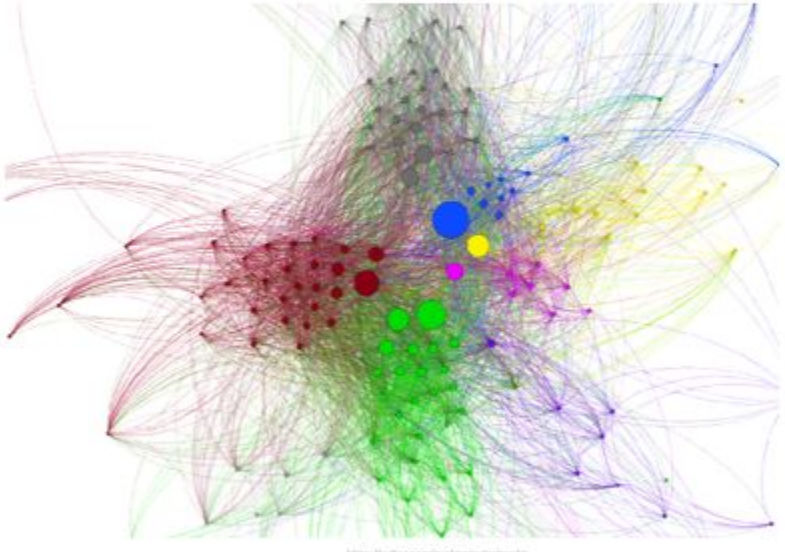
Edward Tufte



Stephen Few

Based on Slide by Michael Sedlmair

But How to Scale to Bigger Datasets?

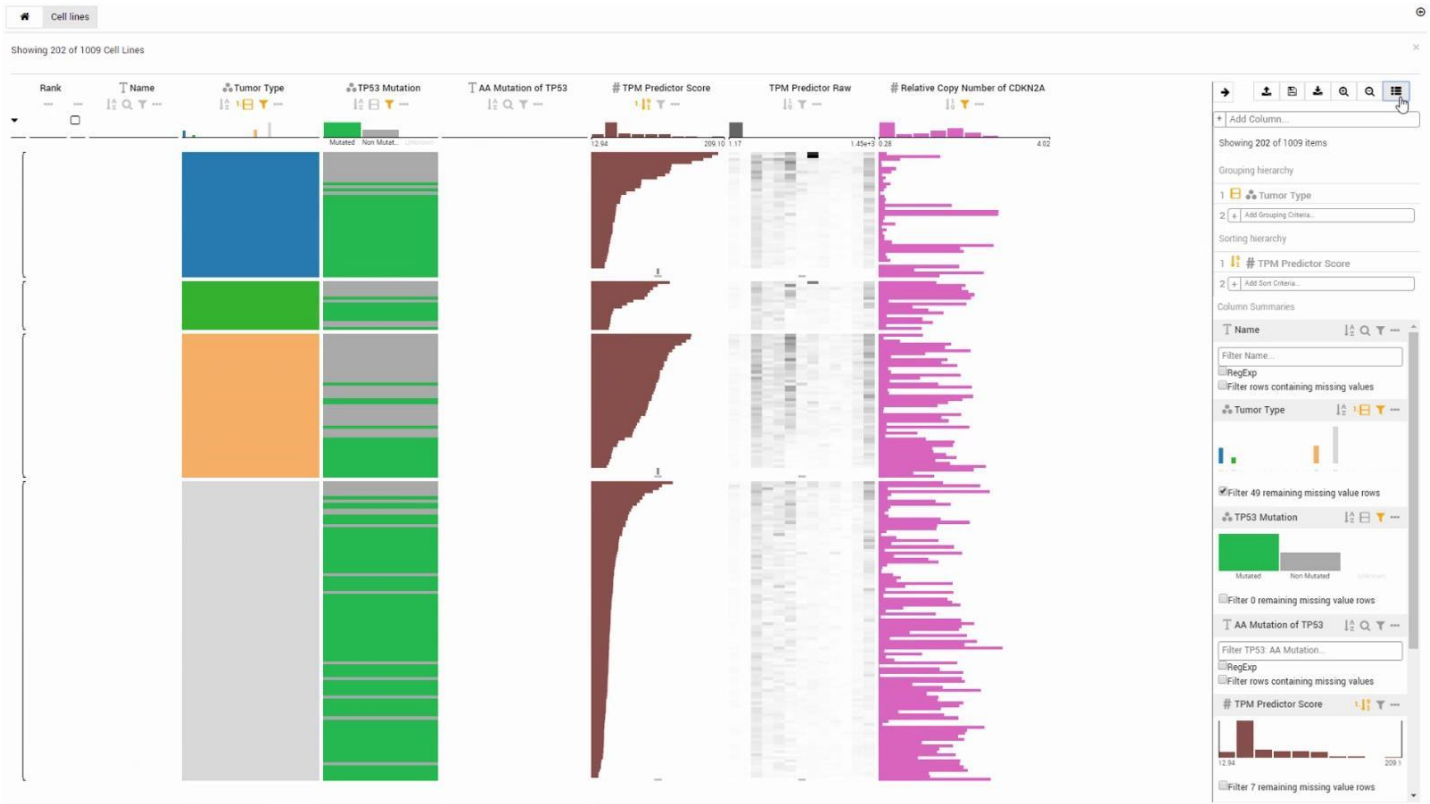


Show all the data?



Hairball Problem

Solution 1: Interaction

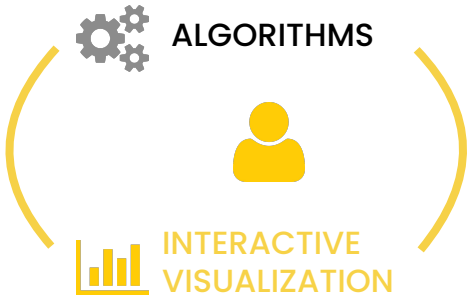


Solution 2: Combining Algorithms and Vis



JOHN TUKEY
(1916–2000)

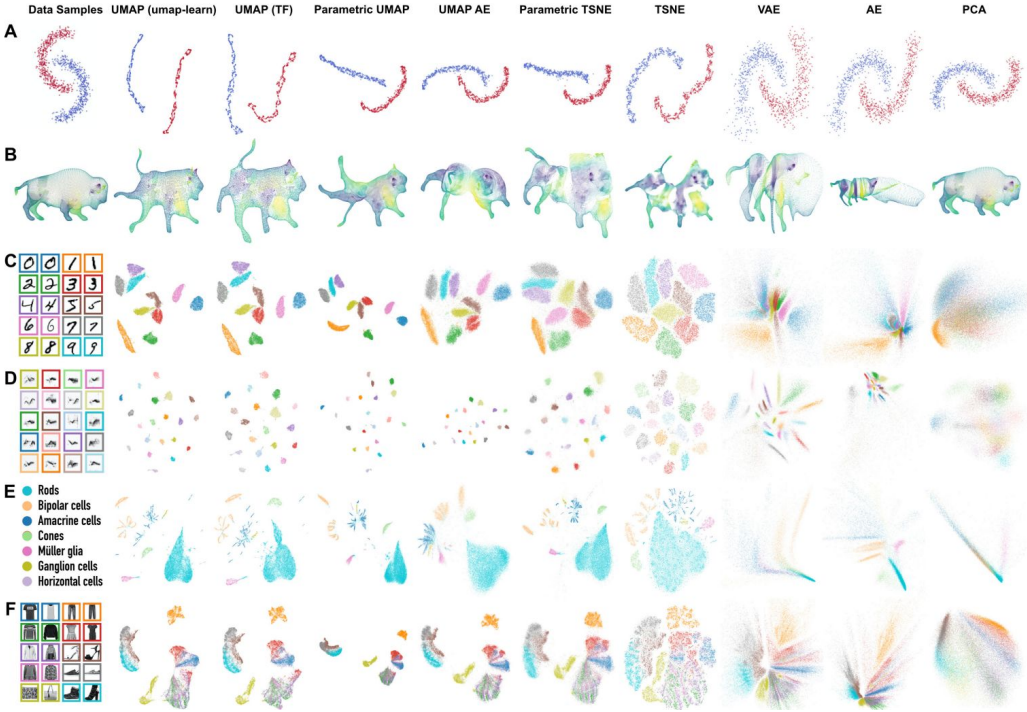
VISUAL ANALYTICS
EXPLORATORY
VISUAL DATA SCIENCE



“THE GREATEST VALUE OF A PICTURE IS WHEN IT FORCES US TO NOTICE WHAT WE NEVER EXPECTED TO SEE.”

Why Algorithms/ML/AI?

Algorithms can reveal patterns that humans cannot see

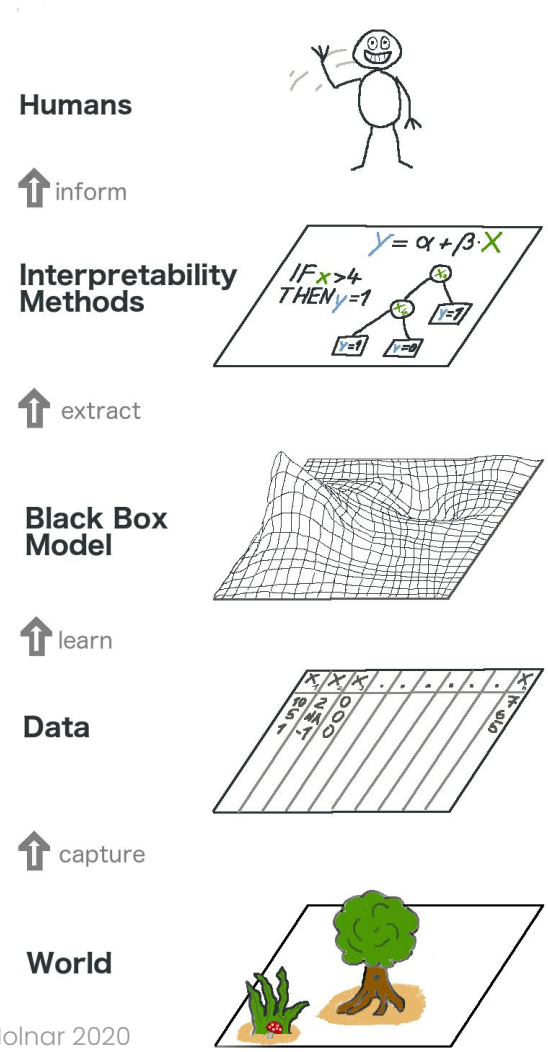


Parametric UMAP, Sainburg et al. 2020

The Big Picture of Explainable AI

Various Purposes

- Interpretability & Explainability
- Debugging & Improving Models
- Comparing & Selecting Models
- Teaching & Understanding Concepts



Molnar 2020

‘Inmates Running the Asylum’ Problem

- XAI used to be done by ML researchers only
- Algorithm-centered XAI often criticized by other scientific fields
- XAI solutions developed often based on researchers’ intuition only

  Better

  - Design XAI solutions for needs of their intended audience
  - Consider how various users interpret and react to explanations depends on the motivational, social, cognitive, professional and educational profiles

WHO is the Human in XAI?

Cater to diverse types of users and stakeholders

- ML experts = model developer
- Data scientist = model user
- Decision-makers using AI systems
  - e.g., doctors, politicians, employment office
- People who might be affected by AI
  - e.g., patients, loan applicant, drivers
- Regulatory agencies
  - e.g., FDA

XAI Questions

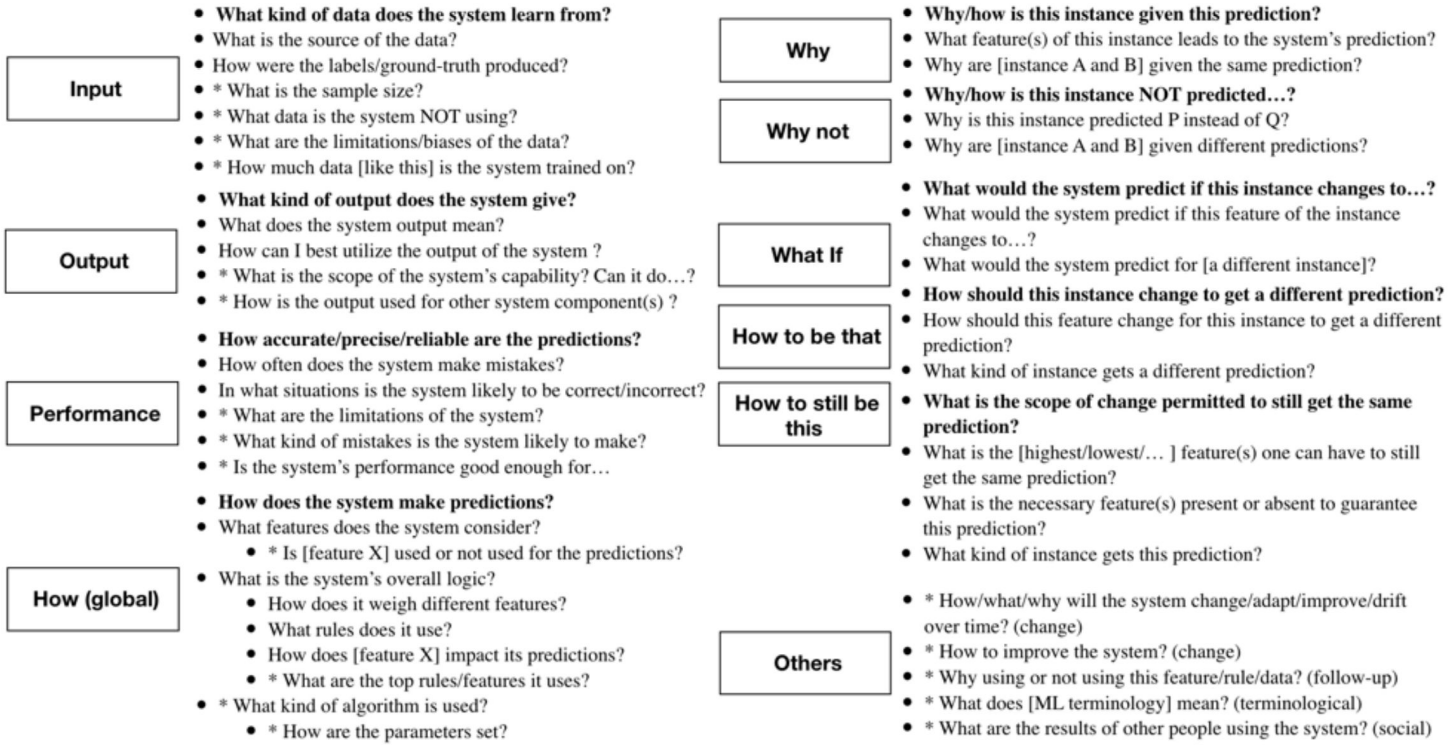


Figure 1. XAI Question Bank, with leading questions in bold, and new questions identified from the interviews with *

Liao et al. 2020 (IBM Research), https://arxiv.org/abs/2001.02478

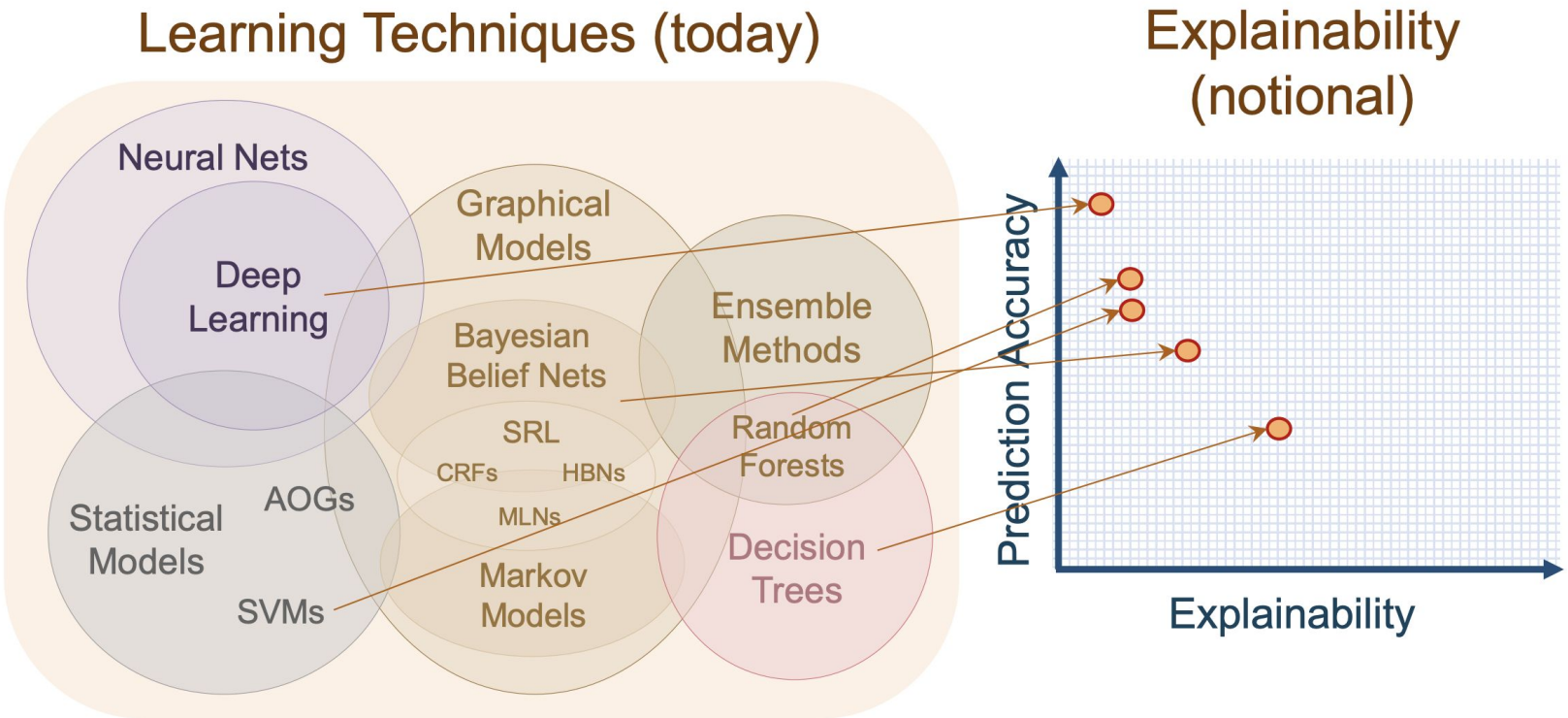
Taxonomy of XAI Methods Mapping to User Questions

Category of Methods	Explanation Method	Definition	Algorithm Examples	Question Type
Explain the model (**Global**)	Global feature importance	Describe the weights of features used by the model (including visualization that shows the weights of features)	[41, 60, 69, 90]	**How**
	Decision tree approximation	Approximate the model to an interpretable decision-tree	[11, 47, 52]	**How**, Why, Why not, What if
	Rule extraction	Approximate the model to a set of rules, e.g., if-then rules	[26, 93, 102]	**How**, Why, Why not, What if
Explain a prediction (**Local**)	Local feature importance and saliency method	Show how features of the instance contribute to the model's prediction (including causes in parts of an image or text)	[61, 74, 83, 85, 101]	**Why**
	Local rules or trees	Describe the rules or a decision-tree path that the instance fits to guarantee the prediction	[39, 75, 99]	**Why**, **How to still be this**
Inspect counterfactual	Feature influence or relevance method	Show how the prediction changes corresponding to changes of a feature (often in a visualization format)	[8, 33, 36, 51]	**What if**, How to be that, How to still be this
	Contrastive or counterfactual features	Describe the feature(s) that will change the prediction if perturbed, absent or present	[27, 91, 100]	**Why**, **Why not**, **How to be that**
Example based	Prototypical or representative examples	Provide example(s) similar to the instance and with the same record as the prediction	[13, 48, 50]	**Why**, How to still be this
	Counterfactual example	Provide example(s) with small differences from the instance but with a different record from the prediction	[37, 55, 66]	**Why**, **Why not**, How to be that

Table 1. Taxonomy of XAI methods mapping to user question types. Questions in bold are the primary ones that the XAI method addresses. Questions in regular font are ones that only a subset of cases the XAI method can address. For example, while a global decision tree approximation can potentially answer *Why, Why not, and What if* questions for individual instances [58], the approximation may not cover certain instances.

Liao et al. 2020, https://arxiv.org/abs/2001.02478

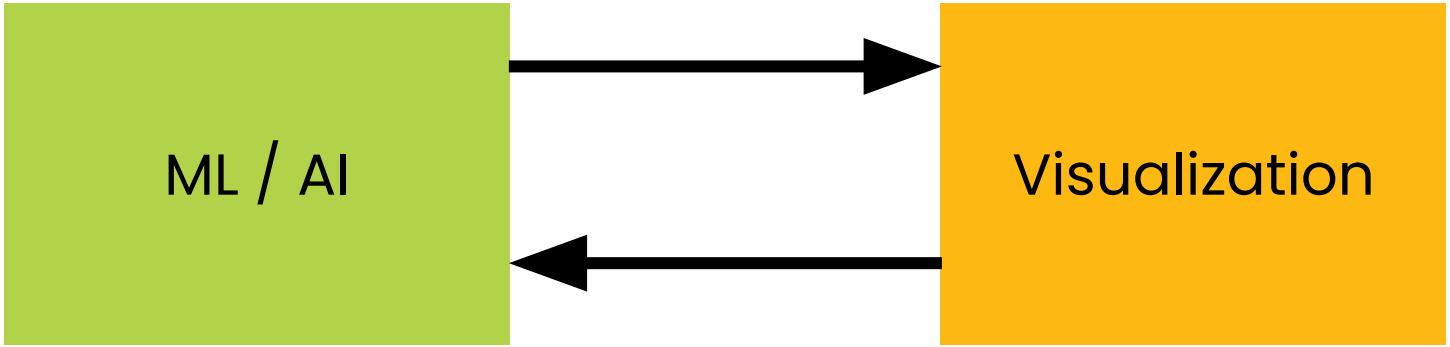
Performance vs. Explainability Tradeoff for ML Techniques



Vis 4 ML vs. ML 4 Vis

ML 4 Vis

Example: Generating visualizations using AI

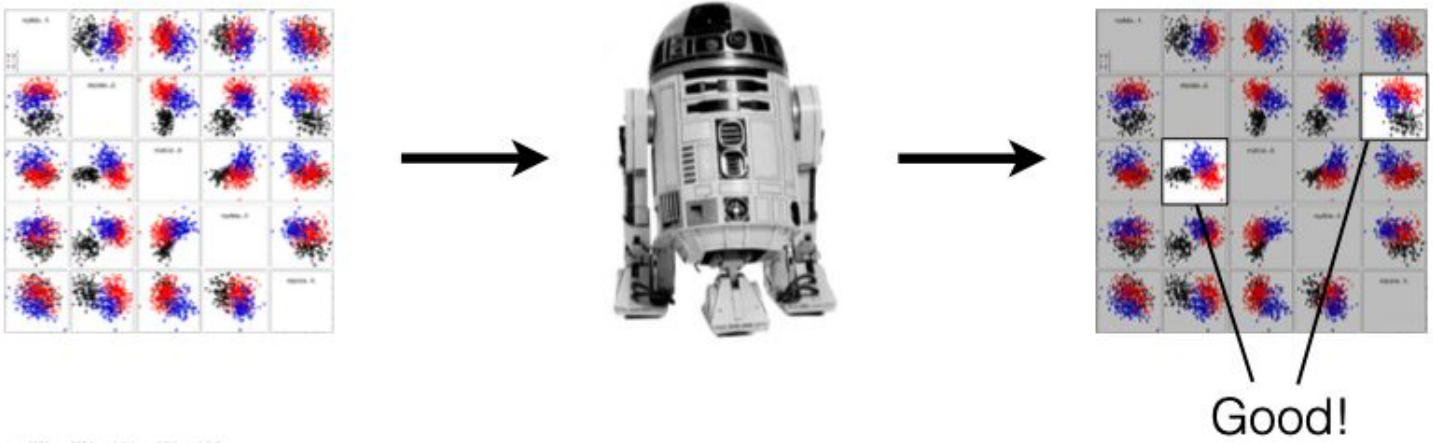


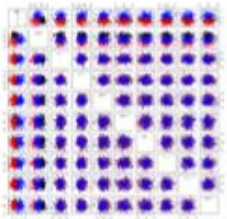
Vis 4 ML = Explainable AI (XAI)

Example: Visualizing latent space of a neural network

ML/AI 4 Vis Usage Example

Help humans finding the relevant visualizations





10D —> 45 Views

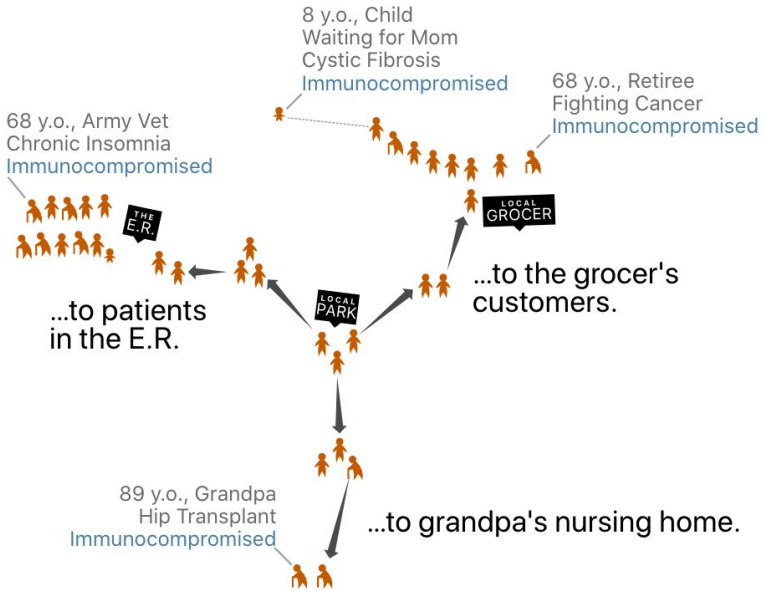
200D —> ~20k Views

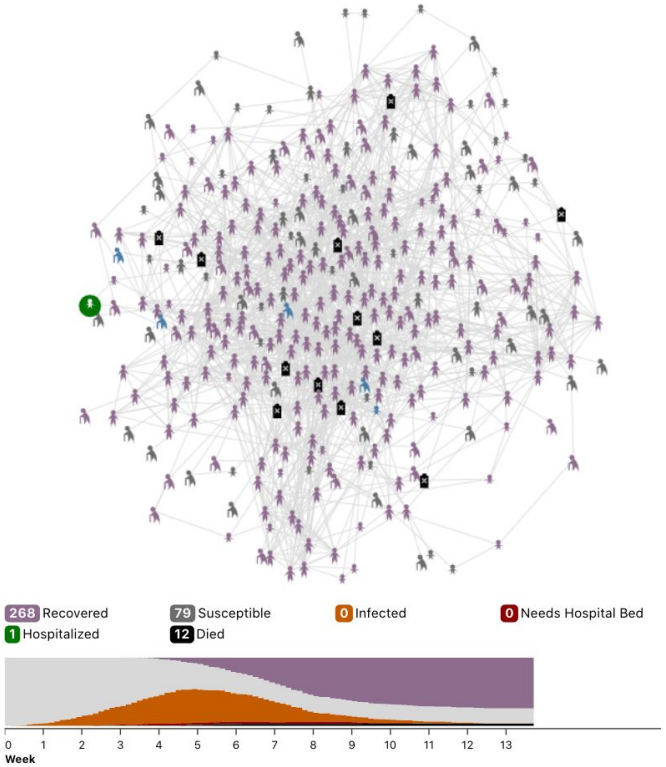
Explaining Algorithms and Models

Example: Making Sense of COVID19 Through Simulations

https://r2d3.us/covid-19/

Scrollytelling Example

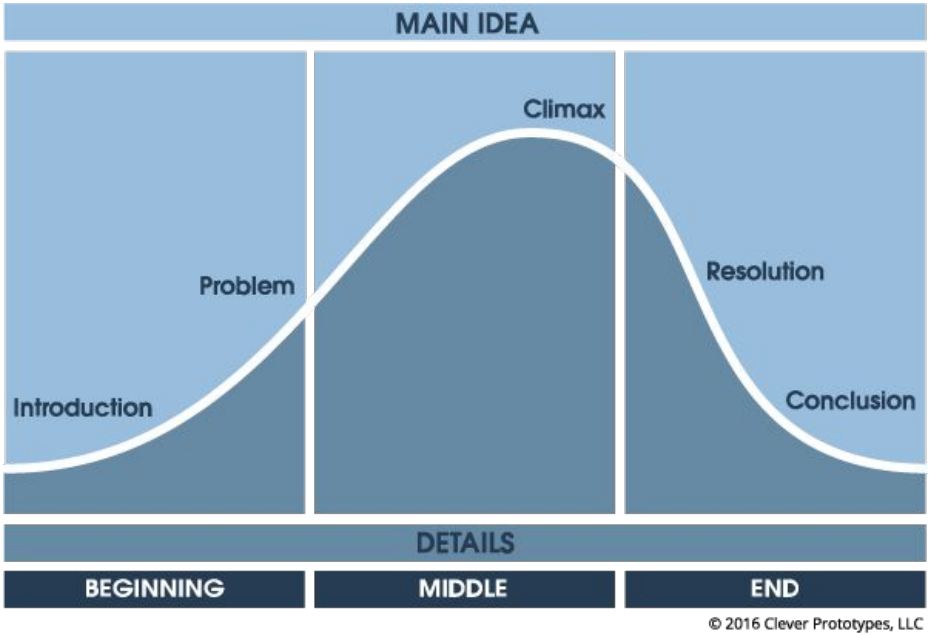




Vis Excursion: Storytelling

- “The world’s second-oldest profession”
- Good stories do more than provide facts and data
  - they situate and give context
  - they engage
  - they educate

- Who / What / Where / When / Why / How



Vis Excursion: Author vs. Reader Driven Storytelling

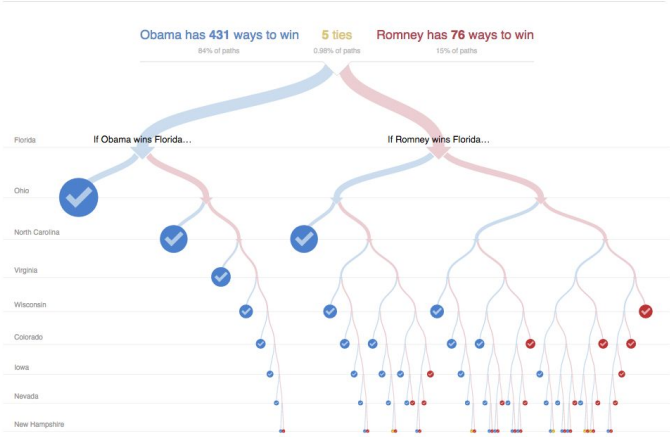
Author Driven ⟷ Reader Driven

Linear Ordering
Heavy Messaging
No Interactivity

No Ordering
No Messaging
Free Interactivity



Al Gore's CO2 Emission Chart

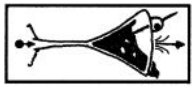


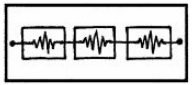
NYT: Paths to the White House

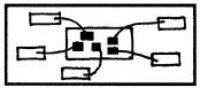
Based on Slide by Alex Lex

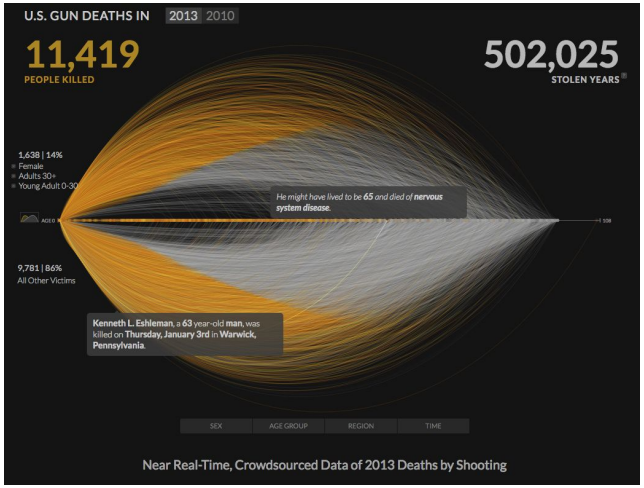
Author vs Reader Driven

- Martini Glass Structure
  - Start with author driven, open up for exploration
- Interactive Slideshow
  - Split into multiple scenes, allow interaction mid-way
- Drill-Down Story
  - Let reader decide which path to follow, all paths are annotated!









http://guns.periscopic.com

What does the COVID example use?

Explaining Sorting Algorithms

Example: Explaining Bubblesort Algorithm

Using Text? —> Not really.

Bubble sort, sometimes referred to as sinking sort, is a simple sorting algorithm that repeatedly steps through the list, compares adjacent elements and swaps them if they are in the wrong order. The pass through the list is repeated until the list is sorted.

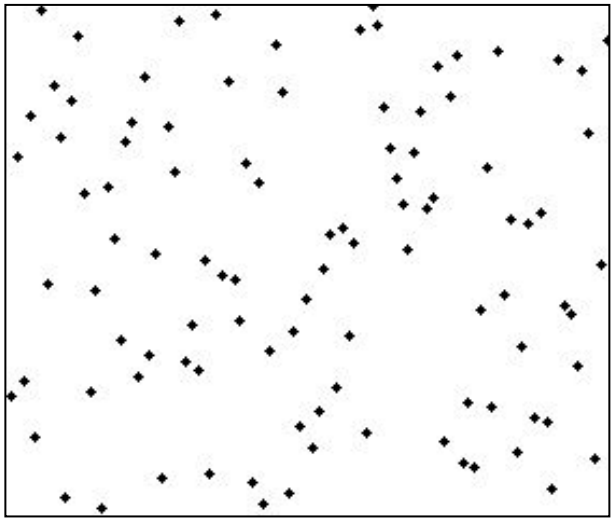
Using Code? —> Not really.

```
procedure bubbleSort(A : list of sortable items)
    n := length(A)
    repeat
        swapped := false
        for i := 1 to n-1 inclusive do
            /* if this pair is out of order */
            if A[i-1] > A[i] then
                /* swap them and remember something changed */
                swap(A[i-1], A[i])
                swapped := true
            end if
        end for
    until not swapped
end procedure
```

Pseudocode implementation

https://en.wikipedia.org/wiki/Bubble_sort

Example: Explaining Bubblesort Algorithm



Try to answer the following questions:

- After what percentage of time is half of the array sorted?
- Can you find an element that moved about half the length of the array to reach its final destination?
- What percentage of the array was sorted after 80% of the sorting process? How about 20%?
- Does the number of sorted elements grow linearly or non-linearly with time (i.e. logarithmically or exponentially)?

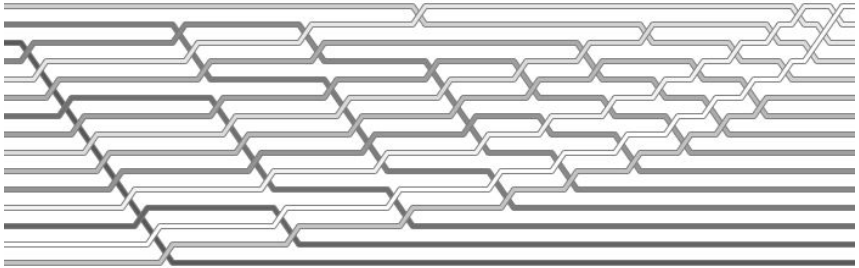
Visualization from Wikipedia

A bubble sort, a sorting algorithm that continuously steps through a list, swapping items until they appear in the correct order.
The list was plotted in a Cartesian coordinate system, with each point (*x*, *y*) indicating that the value *y* is stored at index *x*. Then the list would be sorted by bubble sort according to every pixel's value.
Note that the largest end gets sorted first, with smaller elements taking longer to move to their correct positions.

https://en.wikipedia.org/wiki/Bubble_sort

Example: Explaining Bubblesort Algorithm

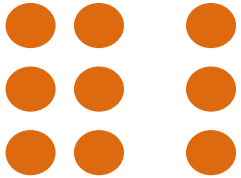
Better: Bump Chart



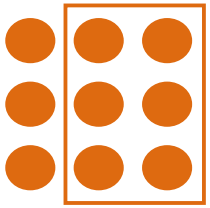
Time progresses from left to right
Brightness indicates value

https://corte.si/posts/code/visualisingsorting/

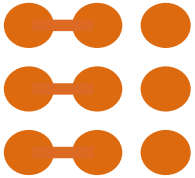
Vis Excursion: Grouping Principles



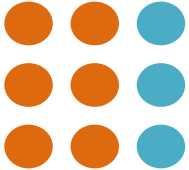
Proximity



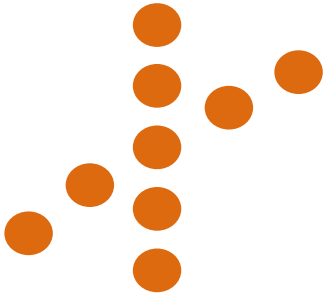
Containment
(Common Region)



Connection



Similarity



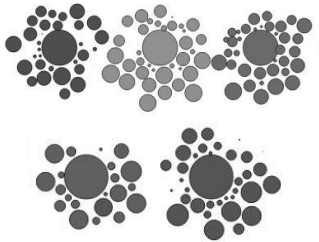
Continuation



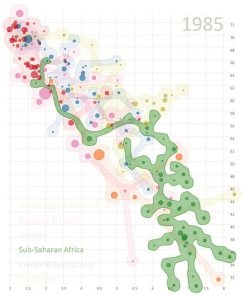
Common Fate



Closure



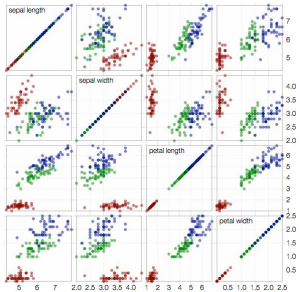
D3.js Example



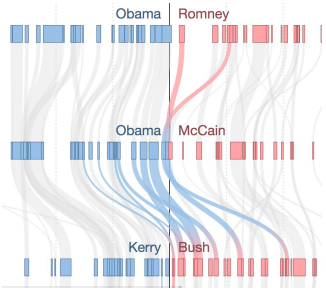
Collins et al. 2009



D3.js Example



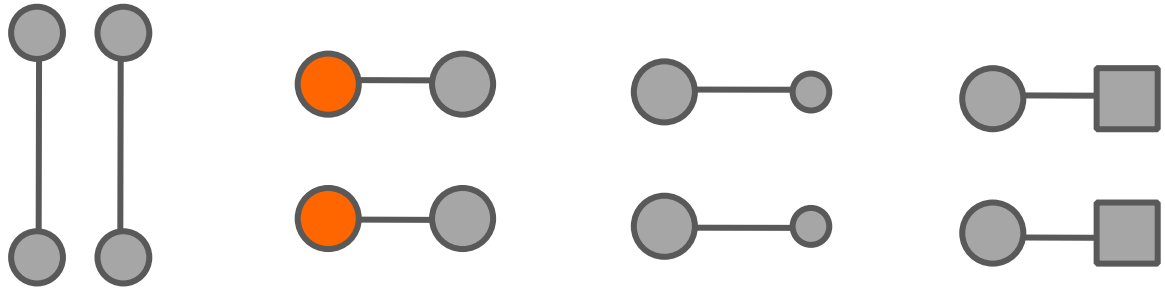
D3.js Example



NYT Swing States

Go back to Gestalt School of Psychology, 1912

Vis Excursion: Grouping Principles

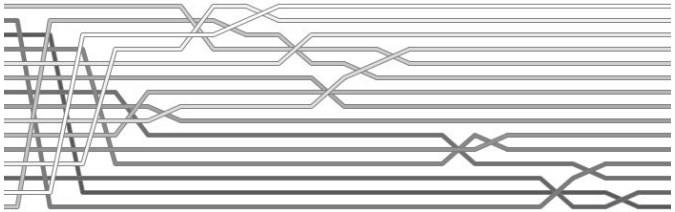


Proximity Color Size Shape

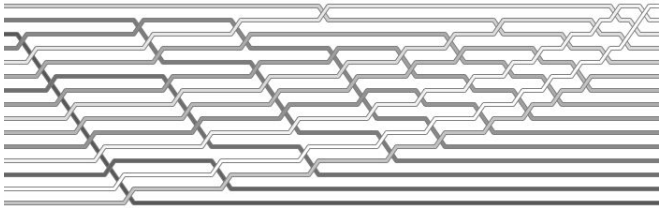
Connection is a very strong grouping principle.
Can overrule others.

Which Grouping Principle Do Bump Charts Use?

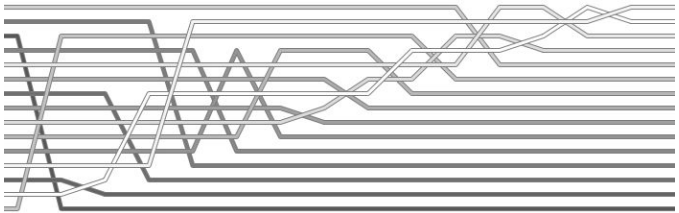
Quicksort



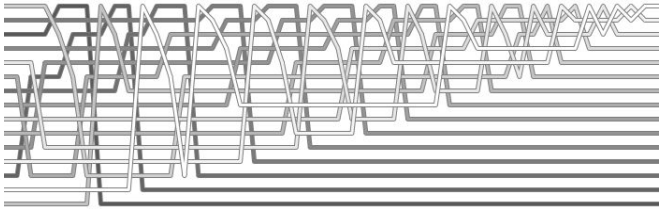
Bubblesort



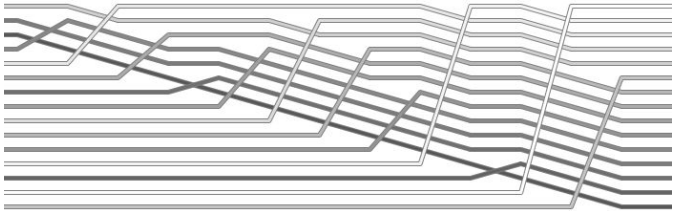
Selection Sort



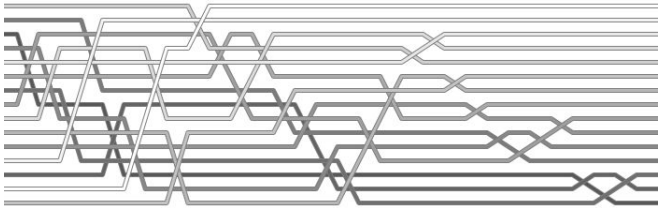
Heapsort



Insertion Sort

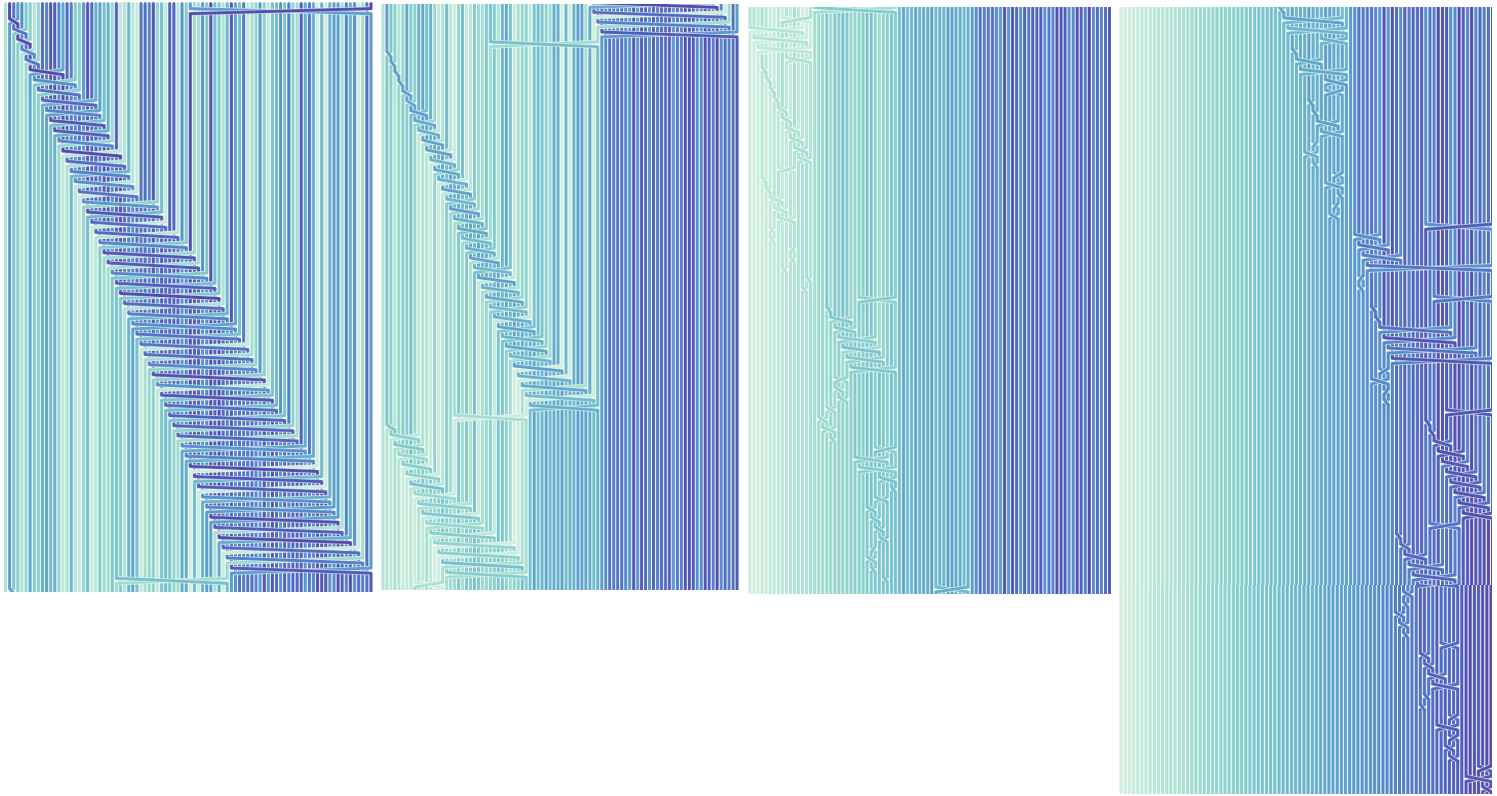


Shell Sort



https://corte.si/posts/code/visualisingsorting/

Example: Explaining Quicksort Algorithm



http://bost.ocks.org/mike/algorithms/

Example: Explaining Quicksort Algorithm

```
function quicksort(array, left, right) {
  if (left < right - 1) {
    var pivot = left + right >> 1;
    pivot = partition(array, left, right, pivot);
    quicksort(array, left, pivot);
    quicksort(array, pivot + 1, right);
  }
}

function partition(array, left, right, pivot) {
  var pivotValue = array[pivot];
  swap(array, pivot, --right);
  for (var i = left; i < right; ++i) {
    if (array[i] < pivotValue) {
      swap(array, i, left++);
    }
  }
  swap(array, left, right);
  return left;
}
```

Quicksort is a [divide and conquer algorithm](). It first divides the input array into two smaller sub-arrays: the low elements and the high elements. It then recursively sorts the sub-arrays. (Wikipedia)

Using Animation: Pivot element = red

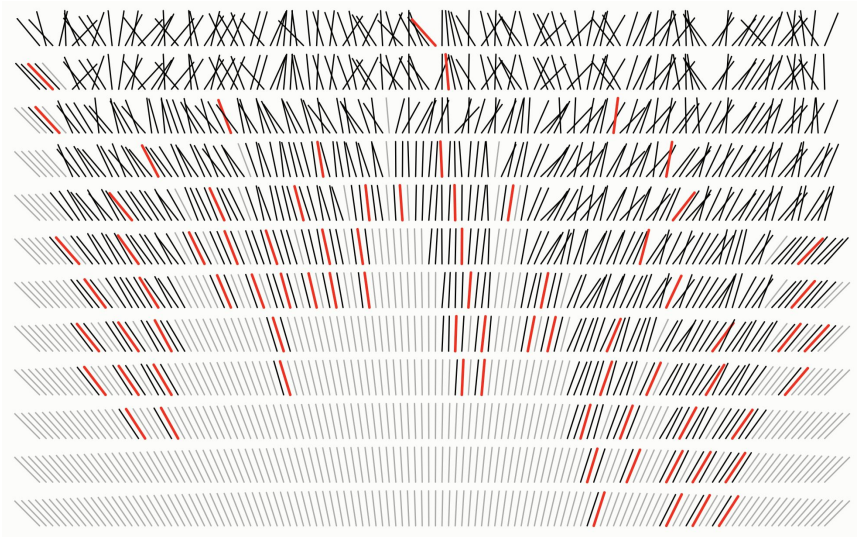


▶ Play

http://bost.ocks.org/mike/algorithms/

Example: Explaining Quicksort Algorithm

Visualization Principle: Eyes beat Memory (= Animation)!



Quicksort

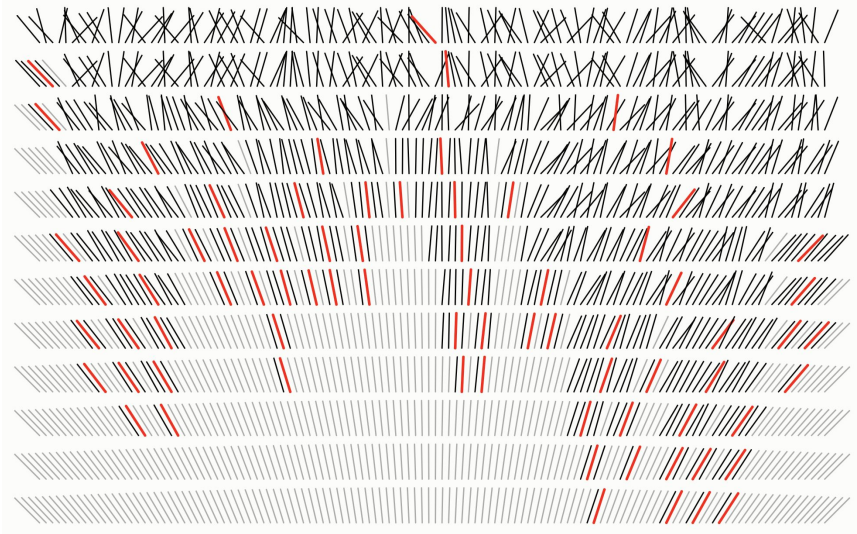
A simple way of turning an animation into a static display is to pick key frames from the animation and display those sequentially, like a comic strip. If we then remove redundant information across key frames, we use space more efficiently. A denser display may require more study to understand, but is faster to scan since the eye travels less.

Below, each row shows the state of the array prior to recursion. The first row is the initial state of the array, the second row is the array after the first partition operation, the third row is after the first partition's left and right parts are again partitioned, etc. In effect, this is breadth-first quicksort, where the partition operation on both left and right proceeds in parallel.

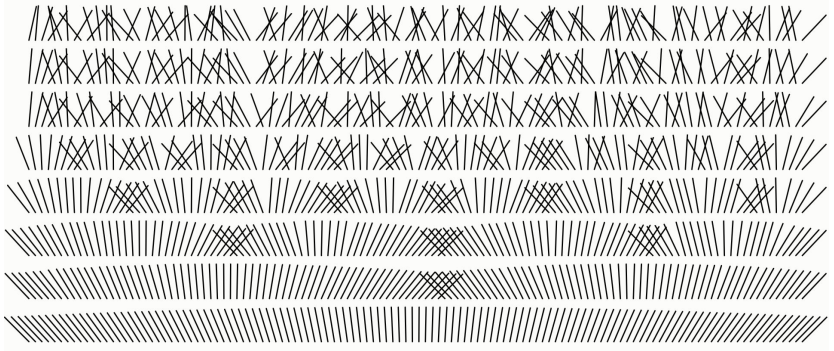
http://bost.ocks.org/mike/algorithms/

Quicksort vs. Mergesort

Visualization Principle: Eyes beat Memory (= Animation)!



Quicksort

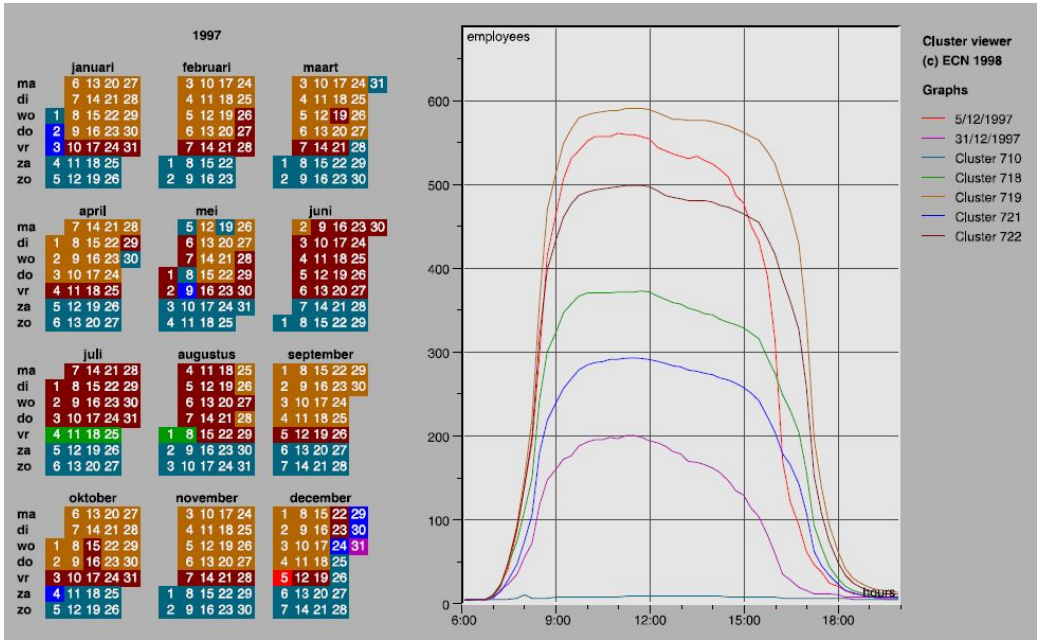


Mergesort

http://bost.ocks.org/mike/algorithms/

Explaining Clustering Algorithms & Results

Example: Visualizing Clustering Results



Visual analysis of the number of employees at work. Day patterns for selected days and clusters are visualized as line plots (right). Individual days in a calendar display (left) are colored according to cluster affiliation.

van Wijk and van Selow 1999

Explaining K-Means Clustering

- Using Text? Quite abstract.

- The K-Means algorithm for clustering basically consists of 5 steps:

  Step 1: Choose the number K of clusters

  Step 2: Select random K points as cluster centers

  Step 3: Assign each data point to the nearest centroid

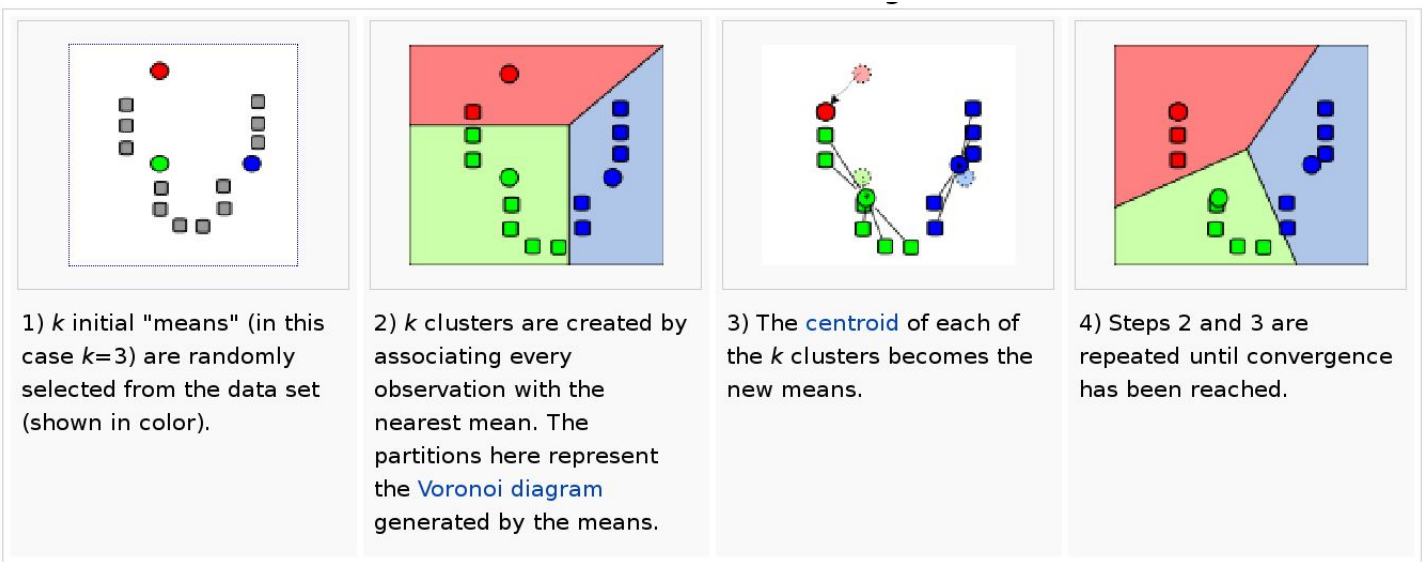
  Step 4: Compute and place the new centroid of each cluster

  Step 5: Repeat step 4 until no observations change cluster

Link to Article by Mikkel Duif 2019

Explaining K-Means Clustering

Partition n observations into k clusters
Each observation belongs to the cluster with the nearest mean



1) *k* initial "means" (in this case *k*=3) are randomly selected from the data set (shown in color).

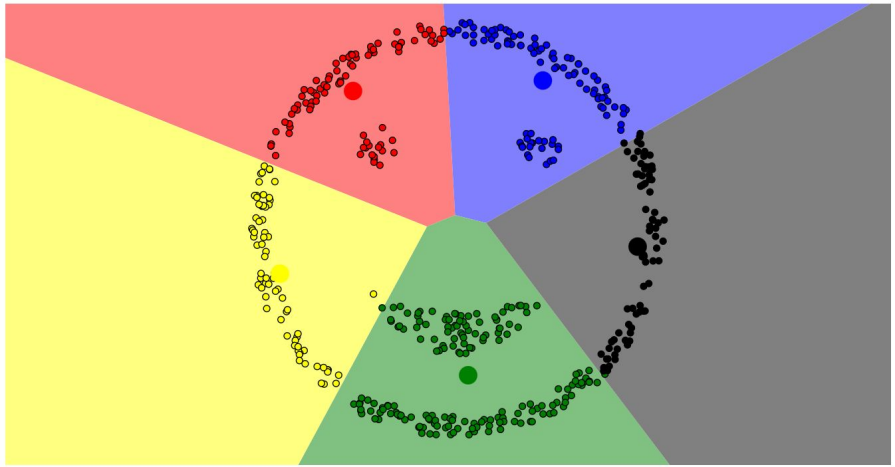
2) *k* clusters are created by associating every observation with the nearest mean. The partitions here represent the Voronoi diagram generated by the means.

3) The centroid of each of the *k* clusters becomes the new means.

4) Steps 2 and 3 are repeated until convergence has been reached.

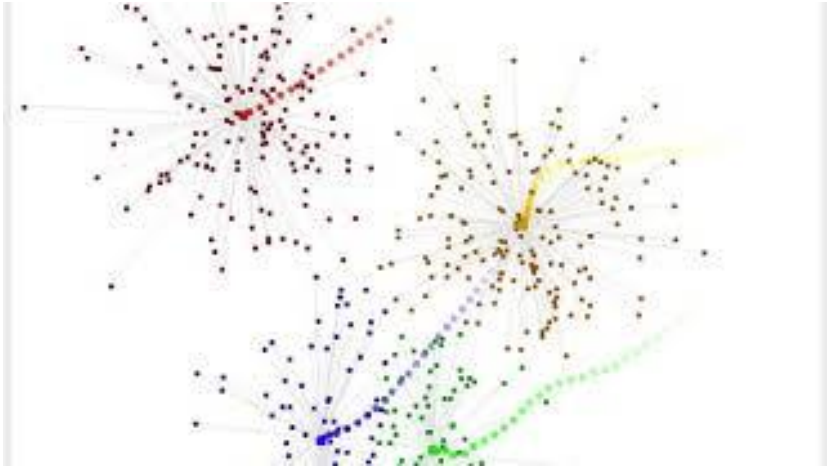
Wikipedia Article on K-Means Clustering

Explaining K-Means Clustering



http://www.naftaliharris.com/blog/visualizing-k-means-clustering/

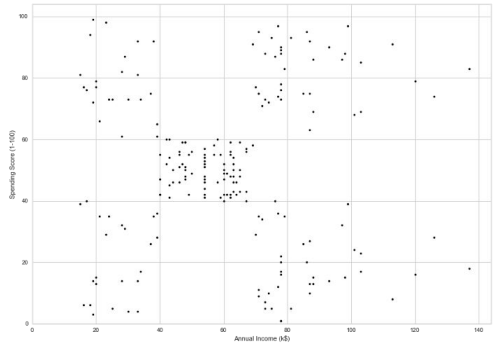
Relies on *Containment & Similarity* Grouping Principles

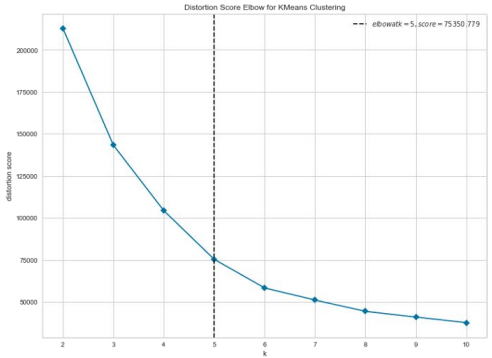


https://www.youtube.com/watch?v=nXY6PxAaOk0

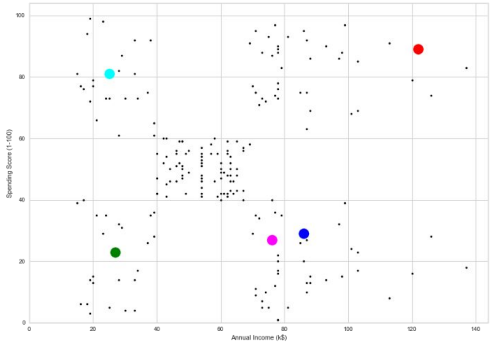
Relies on *Connection* & Similarity Grouping Principles

Towards Data Science Article:
A Visual Introduction to Clustering with KMeans

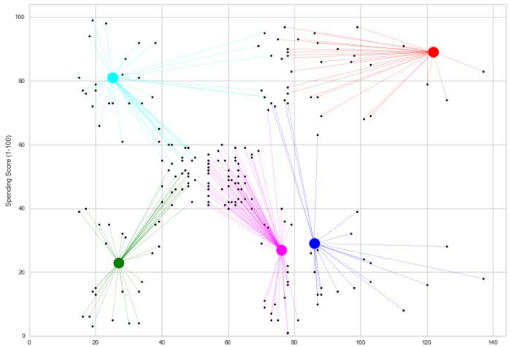




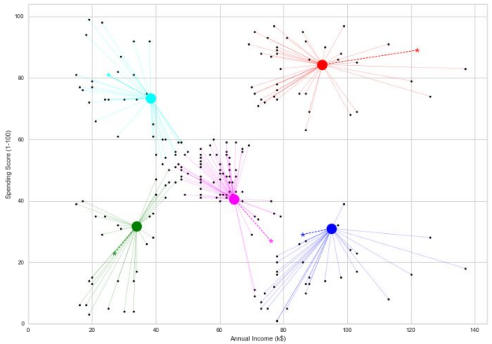
Step 1: Choose the number K of clusters



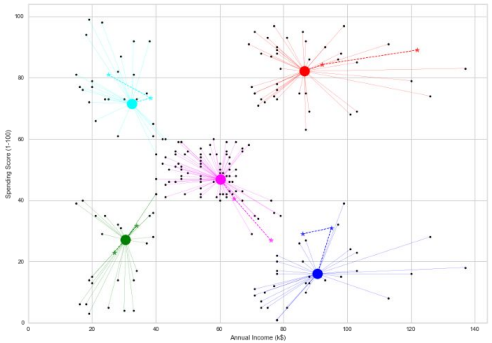
Step 2: Select random K points as cluster centers



Step 3: Assign each data point to the nearest centroid



Step 4: Compute and place the new centroid of each cluster

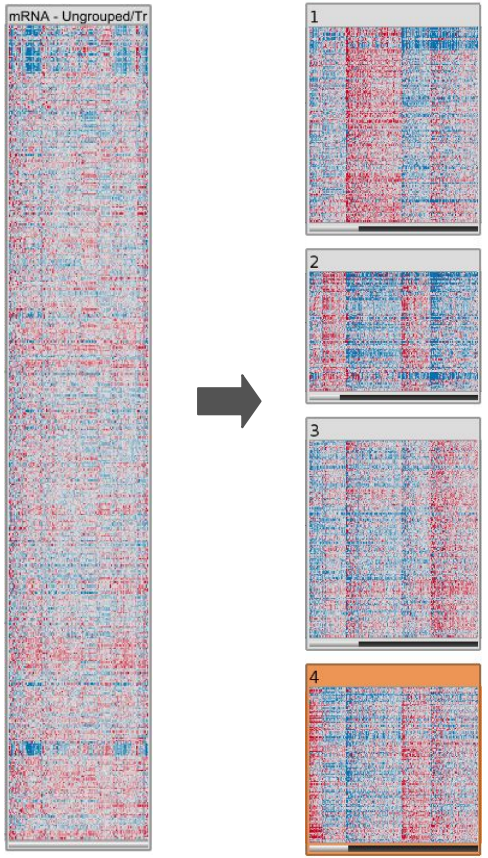


Step 5: Repeat step 4 until no observations change cluster

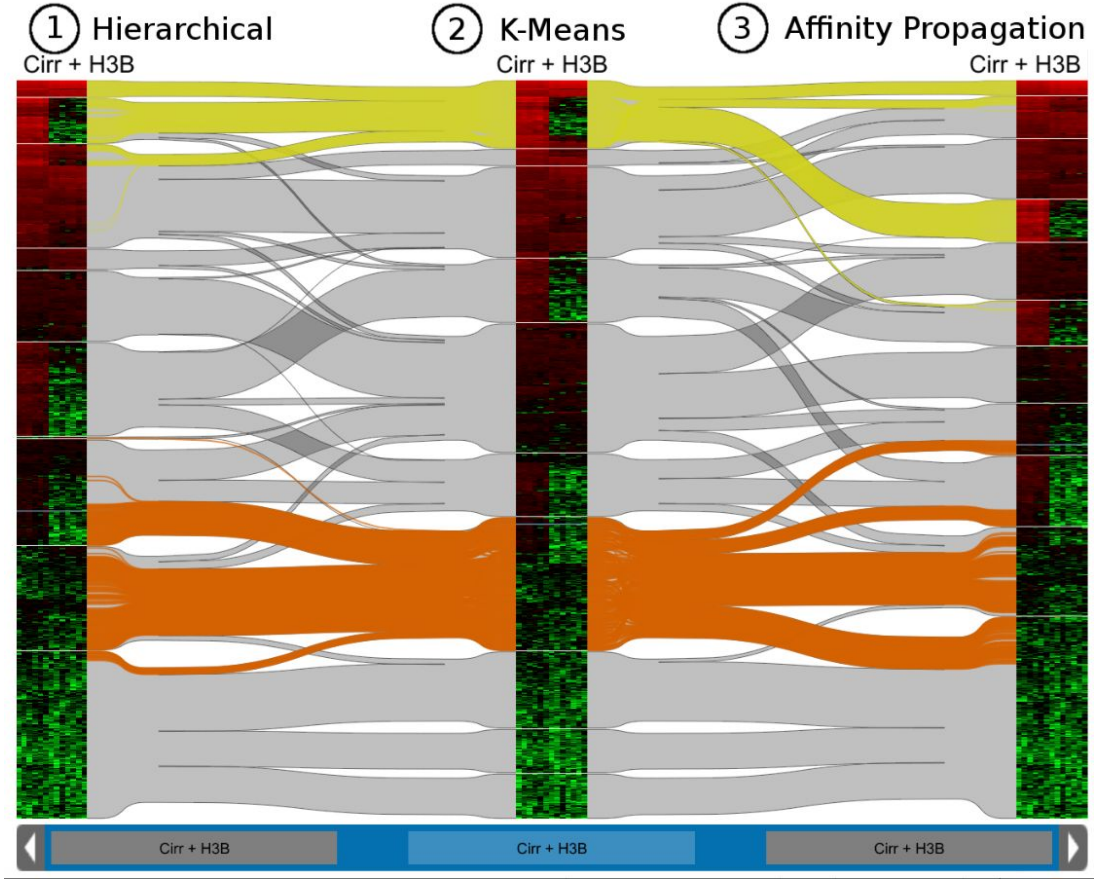
Link to Article by Mikkel Duif 2019

Partitioning Clustering

- Multi-dimensional dataset clustered in one direction of the table
- Results in a binary assignments of items to clusters



Matchmaker Technique for Comparing Clustering Results

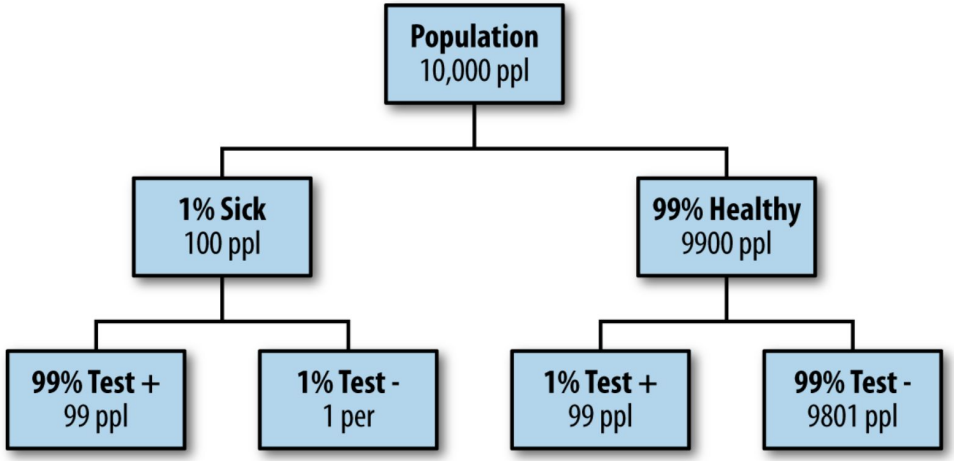


Lex et al. 2010

Explaining Bayes' Theorem

Conditional Probability: Bayes' Theorem

Describes probability of event based on conditions



$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

A, B = events

$P(A|B)$ = probability of A given B is true

$P(B|A)$ = probability of B given A is true

$P(A), P(B)$ = the independent probabilities of A and B

Given that a test is positive, what is the probability that the patient is actually sick?

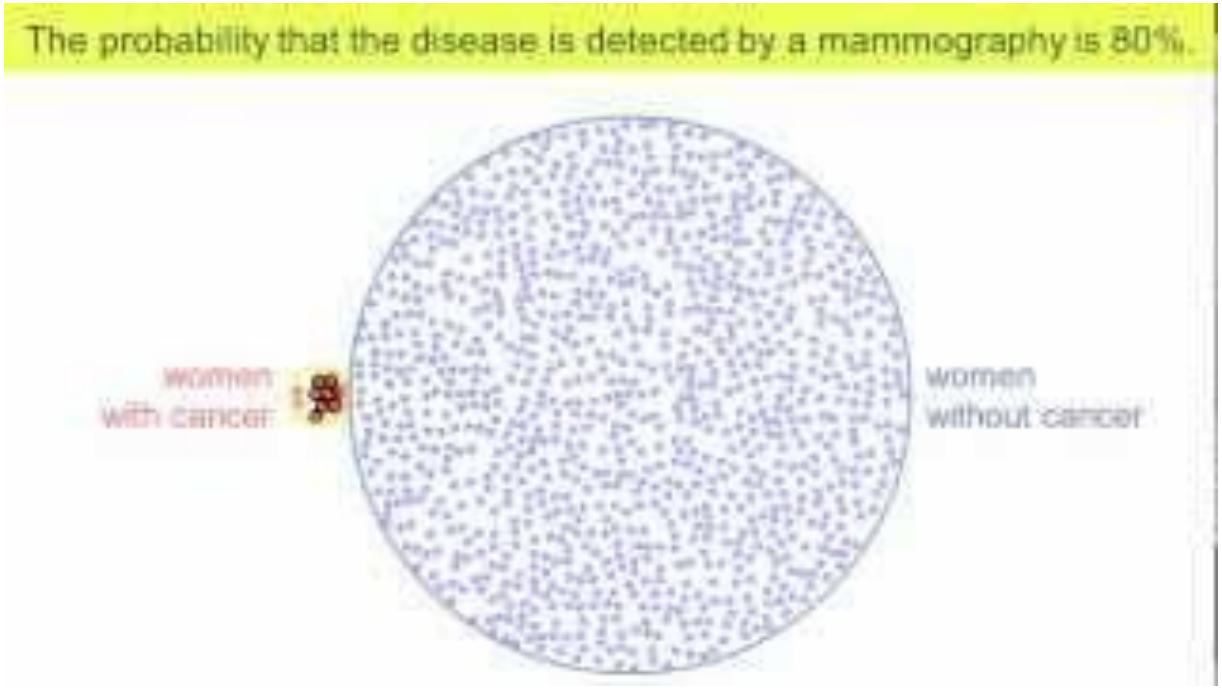
$$p(sick|+) = \frac{p(+|sick)p(sick)}{p(+)} = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.01 \cdot 0.99} = 0.50 = 50\%$$

Sensitivity vs. Specificity

- Sensitivity (true positive rate, recall rate):
  - proportion of positives which are correctly identified as such
- Specificity (true negative rate):
  - proportion of negatives which are correctly identified as such
- Example: Testing for a rare disease
  - 1% of population is infected
  - Goal: Develop highly sensitive and specific test
  - Sensitivity: percentage of sick people who are correctly identified as having the condition
  - Specificity: percentage of healthy people who are correctly identified as not having the condition
  - Given that a test is positive, what is the probability that the patient is actually sick?

Let’s Test Your Intuitive Judgement

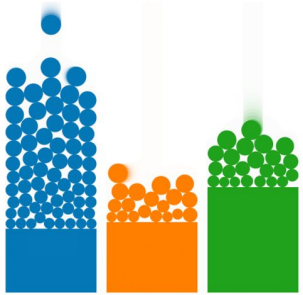
Even for simple problems our intuitive judgement can be wrong.

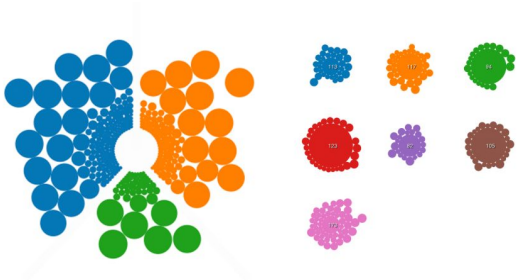


Micallef et al. 2012
https://youtu.be/D8VZqxcu0l0

Vis Excursion: Particle-based Approaches

Idea: Show every data point as a single particle



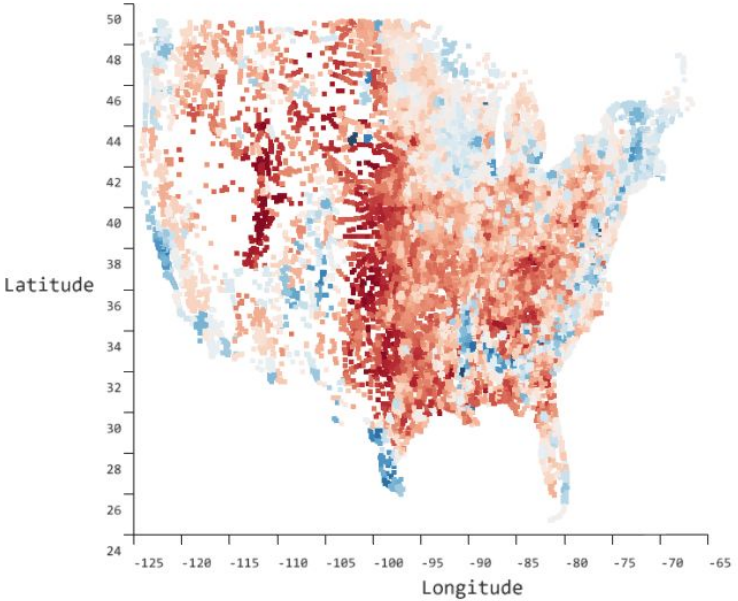






Visual Sedimentation

Huron et al. 2013



Microsoft SandDance
https://microsoft.github.io/SandDance/

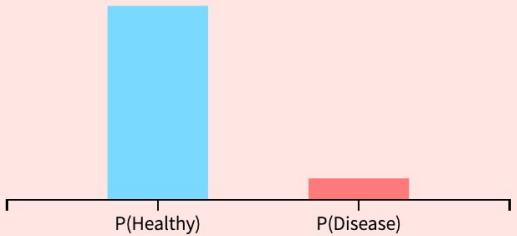
Interactive Explanation from Seeing the Theory

Bayes' Theorem

Suppose that on your most recent visit to the doctor's office, you decide to get tested for a rare disease. If you are unlucky enough to receive a positive result, the logical next question is, "Given the test result, what is the probability that I actually have this disease?" (Medical tests are, after all, not perfectly accurate.) Bayes' Theorem tells us exactly how to compute this probability:

$$P(\text{Disease}|+) = \frac{P(+|\text{Disease})P(\text{Disease})}{P(+)}$$

As the equation indicates, the *posterior* probability of having the disease given that the test was positive depends on the *prior* probability of the disease $P(\text{Disease})$. Think of this as the incidence of the disease in the general population. Set this probability by dragging the bars below.

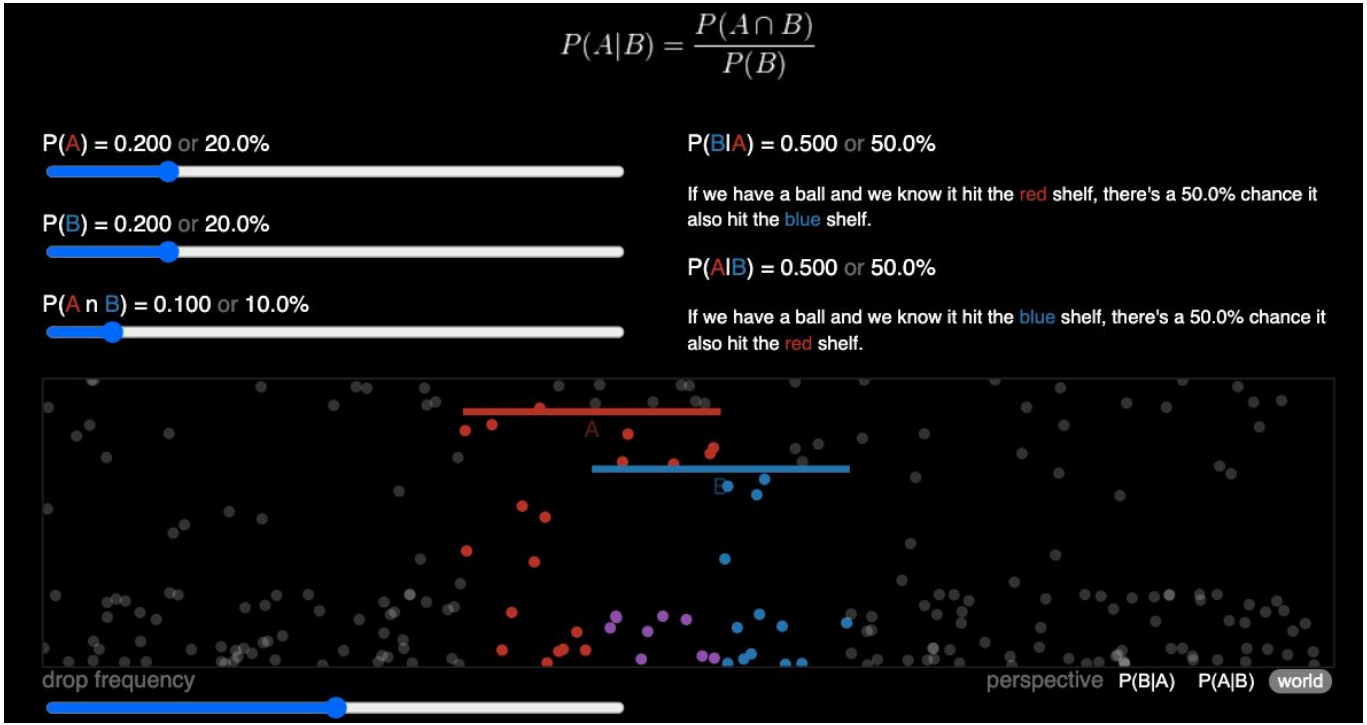


The posterior probability also depends on the test accuracy: How often does the test correctly report a negative result for a healthy patient, and how often does it report a positive result for someone with the disease? Determine these two distributions below.



https://seeing-theory.brown.edu/bayesian-inference/index.html#section1

Visual Explanation of Conditional Probability



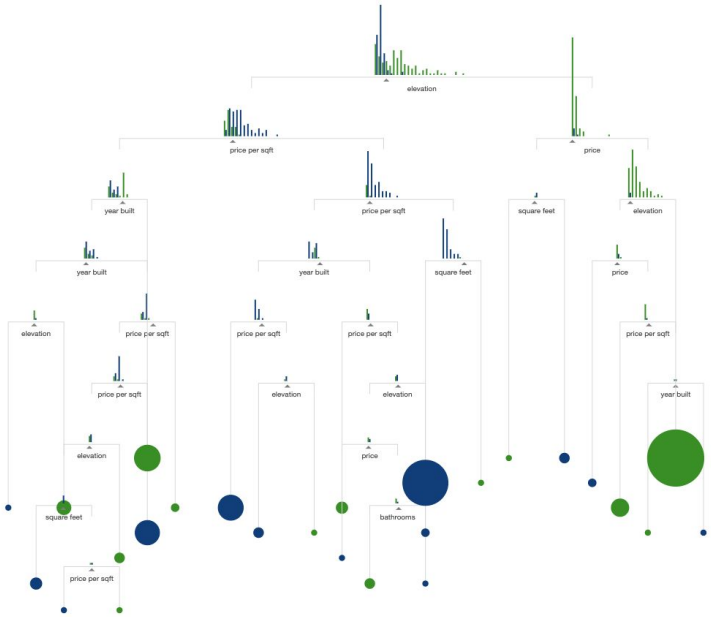
https://setosa.io/conditional/

Explaining Decision Trees

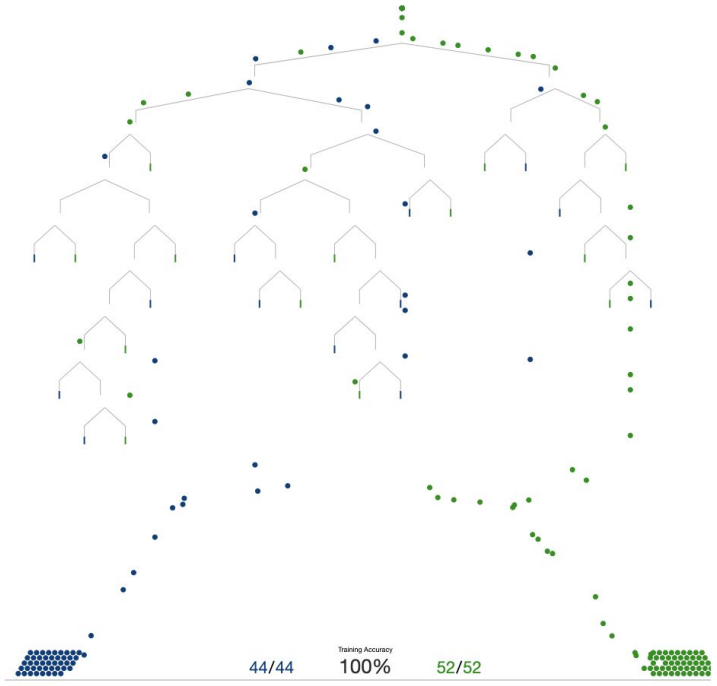
Visual Introduction to Statistical Learning

https://r2d3.us/visual-intro-to-machine-learning-part-1/

Very nice scrollytelling example



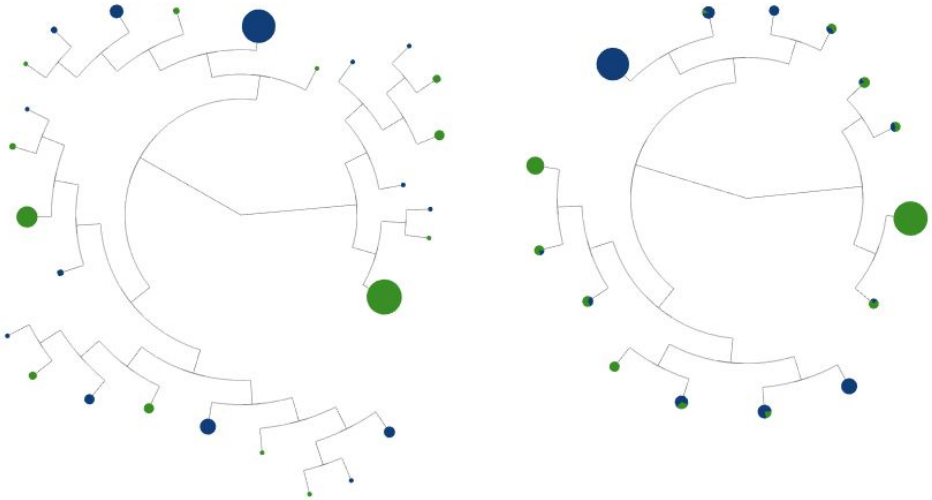
Decision tree model for classifying houses



Particle-based explanation how the decision tree can be used for prediction

Visual Introduction to Statistical Learning

https://r2d3.us/visual-intro-to-machine-learning-part-2/



A VISUAL INTRODUCTION
TO MACHINE LEARNING—PART II

Model Tuning and the Bias-Variance Tradeoff

The goal of modeling is to approximate real-life situations by identifying and encoding patterns in data. Models make mistakes if those patterns are overly simple or overly complex.

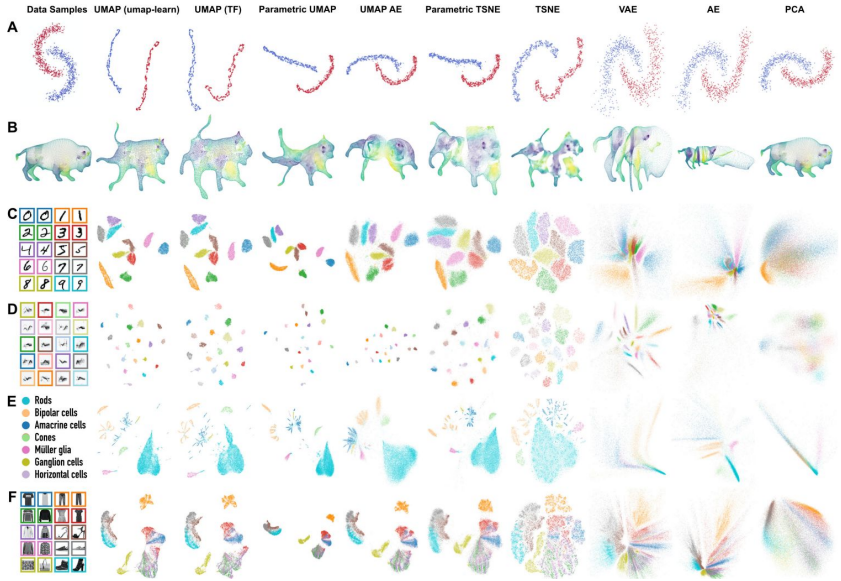
In Part 1, we created a model that distinguishes homes in San Francisco from those in New York. Now, we'll talk about tuning and the Bias-Variance tradeoff.

Summary of Lecture 2

Fundamentals
and
Explaining Algorithms

- AI & VIS Fundamentals

- Explaining Sorting Algorithms
- Explaining Clustering Algorithms & Results
- Explaining Bayes' Theorem
- Explaining Decision Trees



Preview of Lecture 3

Explaining Through Projections

- Dimensionality reduction techniques
  - PCA, UMAP, t-SNE
  - Visualization of embeddings
- Embedding-based trajectories



JKU
Visual Data
Science Lab

AI and Visualization

Univ.-Prof. Dr. Marc Streit

ai-and-vis-course@jku.at

Thank you!